

| Question | Marking Guidance                                                                                                                                                                                           | Mark | Comments                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 05.1     | 1. Starch (solution) in first column;<br>2. Headings for starch concentration/solution <b>and</b> time for (starch) hydrolysis/digestion with mg dm <sup>-3</sup> <b>and</b> minutes/mins/min/m/seconds/s; | 2    | Ignore columns with replicates/mean<br>Ignore lines<br>2. Accept brackets for solidus<br>2. Ignore reference to enzyme<br>2. Reject graph |
| 05.2     | 1. As starch concentration increases, time to hydrolyse/digest starch increases;                                                                                                                           | 1    | Accept converse                                                                                                                           |
| 05.3     | 1. Inhibitor similar shape to substrate;<br>2. Fits/binds to active site;<br>3. Prevents/reduces <u>enzyme-substrate</u> complex forming;                                                                  | 3    | 1. Reject same shape<br>1. Accept 'complementary to active site'                                                                          |
| 05.4     | 1. Less hydrolysis of starch;<br>2. (To) maltose;<br>3. (So) less absorption (of glucose)<br><b>OR</b><br>(So) more egested (starch/carbohydrate);                                                         | 3    | 1. Accept no hydrolysis<br>3. Accept description of egestion, eg lost in faeces                                                           |
| TOTAL    |                                                                                                                                                                                                            | 9    |                                                                                                                                           |